

Processed and Ultra-processed Food.
Directly linked to 32 harmful effects to health, higher risk of obesity, heart disease, cancer, type 2 diabetes

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Abstract.

Scientific research shows that on average, human adults consume daily 1.5-2.5 of food. The daily amount of food intake can vary depending on factors such as age, body weight, activity level, and dietary requirements. Generally, the recommended daily calorie intake is 2,000 calories a day for women and 2,500 for men. The unprocessed or minimally processed foods include fruit, vegetables, milk, fish, fresh meat, pulses, eggs, nuts and seeds that have no added ingredients and have been little altered from their natural state. Processed ingredients that are added to other foods to make them palatable are salt, sugar and fats-oils. Ultra-processed foods (UPF) tend to include many additives and extra ingredients, such as preservatives, emulsifiers, sweeteners, and artificial colours and flavours. The products undergo multiple industrial processes to make them tasty and shelf-stable, and contain additives like emulsifiers, colouring agents and chemical flavours. Ultra-processed foods now account for up to 58% of total daily energy intake in some high income-countries. These highly processed foods contain loads of sugar, salt and saturated fat, as well as other ingredients that can be harmful to many organ and systems within the human body. Scientific epidemiological and clinical research results in the last decades have accumulated evidence linking ultra-processed foods with a 21% greater risk of death from any cause, a 55% increased risk of obesity, a 40% increased risk of type 2 diabetes, a 41% increased risk of sleep problems and a 22% increased risk of depression. A recent review (2022) collected data from 48 prospective studies investigating the associations between UPF consumption and health status, chronic diseases, or mortality. Reviewers concluded that UPF consumption is associated with several adverse health effects and pathologies given the prevalence of UPFs in diets globally. Numerous research efforts have focused on better understanding the mechanisms behind UPF's effects on health. These efforts have brought to light the potential central role played by the intestinal microbiota in connecting UPF and host health. The human intestinal microbiota has co-evolved with their host for millennia and now perform core functions, such as food digestion and maturation of host immunity and metabolism. A combination of laboratory-based preclinical research, epidemiological studies, and clinical trials suggest that UPFs affect human health. Research on UPF showed that is directly linked to 32 harmful effects to health, including a higher risk of heart disease, cancer, type 2 diabetes, adverse mental health and early death, according to the world's largest review of its kind. The findings from the first comprehensive umbrella review (2024) of evidence come amid rapidly rising global consumption of UPF. In the UK and US, more than 50% the average diet now consists of ultra-processed food. For some, especially people who are younger, poorer or from disadvantaged areas, a diet comprising as much as 80% UPF is typical. The review presents all these research studies.

Introduction : The pleasure of eating and the human brain

Humans are wired to derive pleasure from eating food. Processed is any food that is altered during preparation to make it more convenient, shelf-stable or flavourful and palatable. Hyperpalatable food combines high levels of saturated fat, sugar, sodium, or carbohydrates to trigger the brain's reward system, encouraging excessive eating. Humans are wired to derive pleasure from eating nutritious food. In fact, for many people, eating ranks among the greatest pleasures in life! Besides making mealtimes an enjoyable human experience, taking pleasure from food also has significant benefits for human health and wellbeing.

What humans eat, when and how much, all are influenced by brain reward mechanisms that generate 'liking' and 'wanting' for foods. Eating healthy and nutritious foods can make human's brain happy. When humans eat, their brain releases Dopamine, a neurotransmitter that is associated with pleasure and reward. Dopamine helps to reinforce behaviours that are beneficial to survival. Studies show that eating good wholesome food can also help to improve mood, concentration, and memory in humans.

The areas in human's brain that help with regulation of eating and hunger, signal dopamine to be released. This creates a sensation of good feelings, and positive reinforcement. This process gives humans the capacity to feel able to continue their daily lives and control their energy needs. After eating human's behaviour gets better, they 'feel good' again. It is part survival, and part sensory when humans eat nutritious food.

Despite the essential role of food in humans' lives, we have little understanding of the way our knowledge about foods is organized in the brain. At birth, human infants exhibit very few food preferences, and do not yet know much about what is edible and what is not. A multisensory learning development will eventually turn young infants into omnivore adults, for whom deciding what to eat becomes an effortful task. Recognizing food constitutes an essential step in this decisional process. Concepts about food are represented in the human brain at an early age.¹

Regularly consuming anything that is too processed, fatty or high in added sugar and salt can increase brain inflammation and may impair human memory, and processed food usually do not contain any beneficial nutrients. Most types of modern food include also ultra-processed ingredients.



The first rule of brain health is: what's good for your heart is good for your brain. The foods that your primary care doctor would encourage you to eat for your best heart health are the same as what maximizes your brain health. This applies to human activities as well: regular exercise is good for heart and brain. When humans make healthy choices, they can feel twice as good about themselves knowing that their heart and mind feel good too. The second rule is that vitamins and supplements are not as effective as lots of people might think.

Research shows that getting the nutrients humans need from their daily food is much more effective than taking dietary supplements. When humans eat whole foods, they get more than just the vitamins and minerals. They get all the complex **micronutrients** that come with wholesome food, including fiber and a series of protective antioxidants substances. These nutrients are also easier for the human body to absorb when they come from food than from a vitamin supplement. The healthy recommendation from doctors is always: get the best nutrients from whole food and talk to your primary care doctor. Do not take dietary supplements without professional medical advice. [Health Partners, The best food for brain health, <https://www.healthpartners.com/blog/food-for-the-brain/#:~:text=The%20fi...>].

Nutritious and proper food ensures survival as a source of pleasure and well-being. In order to survive, human brains must optimize resource allocation for reward pursuit. Humans focus on how wanting, liking and

learning related to human food intake. Eating uses the pleasure-reward physiology to motivate humans to eat. Endogenous opiates, such as morphine, insure our survival by helping humans to make eating motivational via pleasure induction. Eating behaviour is controlled by a complicated system, which includes the central and humoral regulating elements. The key role in ensuring the central mechanisms functioning is played by the cortex and reward zones in the limbic system (hedonic regulation), where the stimuli originating from the environment are analyzed; they include both alimentary (appearance, taste and food smell) and non-alimentary stimuli (emotional discomfort and stress).²

Pleasure and reward are key mechanisms ensuring that individuals and species seek the fundamental rewards allowing survival and procreation; and as such have been proposed as evolution's boldest trick. Human eating behaviour is guided by heat-seeking missiles. Other influences in eating behaviour include cognition, emotion and reward and may lead to eating beyond, or below homeostasis, e.g. obesity or anorexia respectively. A state of balance among all the body systems needed for the body to survive and function correctly. In homeostasis, body levels of acid, blood pressure, blood sugar, electrolytes, energy, hormones, oxygen (O₂), proteins, and temperature are constantly adjusted to respond to changes inside and outside the body, to keep them at a normal level.³

What are processed (PF) and ultra-processed foods (UPFs)?

Some foods are much more processed than others. A box of macaroni and cheese or a microwavable dinner, are considered heavily processed. These foods are called ultra-processed foods because they've been chemically altered with artificial flavours, additives (e.g. salt) and other processed plant or animal ingredients.

We must distinguish between **unprocessed or minimally processed foods**, like fresh fruits and vegetables. Unprocessed foods are natural edible food coming without any alteration from plants and animals. Minimally processed foods have been slightly altered so they can be more easily stored,

prepared, and eaten; but this minimal processing does not substantially change the nutritional content and composition. Examples include cleaning and removing inedible or unwanted parts, grinding, refrigeration, pasteurization, fermentation, freezing, and vacuum-packaging. This allows the food to be preserved for more time and remain safe to eat. Many fresh fruits, vegetables, whole grains, nuts, meats, plain yogurt with no added sugar or artificial sweeteners, fresh and dried pasta, tea, coffee, and milk fall into this category.

Processed foods are generally thought to be packaged food item containing many ingredients, perhaps even artificial colours, flavours, or other chemical additives. These processed foods are often referred to as convenience or pre-prepared foods. Various studies in the past showed that excessive consumption of processed foods contribute to the obesity epidemic (especially in western societies) and the rising prevalence of chronic diseases like heart disease, type 2 diabetes and cancer. However, the definition of processed food varies widely depending on the source:

Also, processed food is considered as one that has undergone any changes to its natural state (washing, milling, cutting, heating, pasteurizing, blanching, cooking, canning, freezing, drying, dehydrating, packaging) or other procedures that alter the food from its natural state. The processed food may include also the addition of other ingredients such as preservatives, flavours, nutrients and other food additive such as salt, sugars, and fats.

Ultra-processed foods and drinks (UPFDs), commonly referred to as “highly processed foods”, are the ones with the incorporation of salt, sugar and fat. Also, artificial colours and flavours, preservatives, antioxidants, thickeners, emulsifiers, and artificial sweeteners that promote shelf stability, preserve and enhance texture, and increase palatability. Several processing steps are involved in an ultra-processed food. They are often mass-produced with low-cost ingredients making them cheap and highly profitable. Ultra-processed foods are typically ready-to-eat with minimal additional preparation. Not all but some of these foods tend to be low in fiber and nutrients. Examples are sugary drinks, cookies, some crackers, chips, and breakfast cereals, some frozen dinners, and luncheon meats. Studies for many years have

shown an association between the increasing consumption of ultra-processed foods and the rise in obesity in Western societies (USA, Canada, EU, etc).⁴

Most foods consumed in western societies in the last decades have been processed or ultra-processed in some way. However, the concern about *processed food* is not over items like canned tomatoes or canned tuna, which are processed to lock in freshness and nutrients. The concern is over more heavily processed foods like crackers, jarred pasta sauces and cake mixes. It is estimated that more than 50% of the calories in the average U.S. American diet come from ultra processed foods. In another country, like Canada, the percentage is slightly lower, at 48%. Most OECD countries have similar consumption of processed and ultra-processed foods.

The energy share from UPFDs (ultra-processed food and drinks) varied markedly across the 22 European countries included, ranging from 14 to 44%, being the lowest in Italy and Romania, while the highest in the UK and Sweden. Fine bakery wares and soft drinks were most frequently ranked as the main contributors. Countries with a higher sugar intake reported also a higher energy share from UPFDs. No associations with fibre intake or high BMI (Body Mass Index) were observed. Population-level UPFDs consumption substantially varied across Europe, although main contributors are similar. UPFDs consumption was not observed to be associated with country-level burden of high BMI, despite being related to a higher total sugar intake.⁵

Previous research using total food/drink volume sales per capita from Euromonitor (including 80 countries) showed that UPFD volume sales is the highest in Western Europe, North America (US, Canada, Mexico), and Australasia (Australia, New Zealand and some neighbouring islands in the Pacific Ocean), while also alarmingly increasing in other world regions. Baked goods, including cakes, pastries, industrial breads, and soft drinks ranked among the top contributors to sales of UPFDs. In Europe, UPFDs contributed to an average of 25% of the total dietary energy with the lowest estimates observed in Portugal and Italy and the highest in Germany and the UK, as estimated from the household budget survey data from 19 European countries. Only a few studies from Belgium, France, Portugal, and the UK have used national representative individual-level dietary data reporting a total proportion

of daily energy intake from UPFDs of 30% -- 55% with the highest share for baked goods and confectionary, processed meats, and soft drinks.⁵

In recent decades, ultra-processed convenient food intake has increased dramatically worldwide. These foods now account for 25–60% of a person's daily energy intake throughout much of the world.⁶

For many decades food experts and medical professionals, have accumulated evidence that high consumption of processed and ultra-processed foods has been linked to *various health issues*:

Nutritional Deficiencies: Ultra-processed foods are often stripped of essential nutrients during processing, often leading to an unbalanced diet that lacks vitamins, minerals, and natural fibre.

Excessive sugar, salt, and poor quality fats: Processed foods tend to be high in added sugars, sodium, and unhealthy fats (saturated or trans), potentially contributing to weight gain (obesity), unbalanced blood sugar levels and gut disturbances.

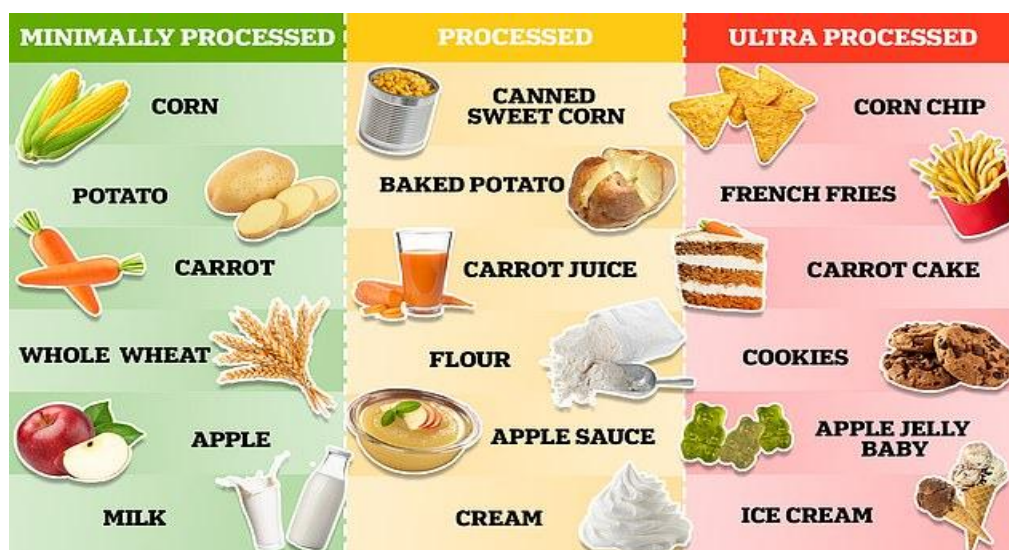


Figure 1. Unprocessed foods would be whole fruits, vegetables, fresh meat and fish, unaltered/raw grains, legumes, and nuts.

Processed food: breakfast cereals, cheese, tinned vegetables, bread, savoury snacks, such as crisps, sausage rolls, pies and pasties, meat products, such as bacon, sausage, ham, salami, microwave meals or ready meals, cakes and biscuits, fizzy drinks.

Ultra-processed foods: are typically stripped of most of their nutrients and include a lot of additives (which, sadly, is part of what makes them so tasty)—from those that you would have trouble pronouncing to common culprits like sugar (in various forms), salt, saturated fat, and trans fats.

Digestive Issues: Processed foods may contain artificial additives and preservatives that can upset the gut's natural balance, which may lead to digestive concerns and dysbiosis. Dysbiosis is an imbalance in bacterial composition, changes in bacterial distribution within the gut. There are 3 types of dysbiosis: a. Loss of beneficial bacteria, b. Overgrowth of potentially pathogenic bacteria, and c. Loss of overall bacterial diversity.

Long-term Health Risks: many epidemiological and clinical studies showed that regular consumption of ultra-processed foods has been associated with an increased risk of obesity, Irritable Bowel Syndrome, type 2 diabetes, and other disorders. Studies have shown that a high intake of ultra-processed food can negatively impact gut health and Inflammatory Bowel Disease.

Ultra-processed and cancer. Eating regularly a greater amount of ultra-processed food was associated with a higher risk of developing cancers of the upper digestive tract, such as mouth, throat and esophageal cancer.

A recent review study (2023) showed that the highest UPFs consumption was associated with increased risk of colorectal cancer (OR = 1.23), colon cancer (OR = 1.25), and breast cancer (OR = 1.10) but not rectal and prostate cancer.

[Lian Y, Wang G-P, Chen G-Q, Chen H-N. Association between ultra-processed foods and risk of cancer: a systematic review and meta-analysis. *Frontiers Nutrition* 10:1175994, 2023. doi: [10.3389/fnut.2023.1175994](https://doi.org/10.3389/fnut.2023.1175994).].

Evolved consumption of processed and ultra-processed food

Since the dawn of humankind, humans have built and used a variety of techniques to process their food and make them palatable. From hunter-gatherer, to pastoral-migrant, to peasant-agricultural lifestyles, these tools to process food have evolved over hundreds of thousands of years. In the modern societies, after the building of towns and cities, humans needed to provide their homes with food, usually obtained from the neighbouring countryside, with sun drying, salting, pickling, and smoking of foods used for preservation when they could not be freshly consumed. Over time, more sophisticated methods were developed, mainly to preserve foods or to exploit cereals (e.g., by use of mills to process flour to be mixed with water afterwards to create wheat bread). During the industrial revolution, processing

technologies were invented, enabling larger scale production of culinary ingredients, such as oils, animal fats, sugars, flour, and salt.^{7,8}

Nearly a century later, due to industrial processing techniques, a wide range of products with guaranteed microbiological safety (very important characteristic of modern food), that are easy to preserve, practical, and convenient to consume in various social contexts, became accessible and affordable to most populations in high-income countries. Mechanical and physical techniques (e.g., roller milling, pressure rendering, and extrusion) and chemical techniques (e.g., hydrogenation and hydroxylation) were developed, through the use of artificial flavours, preservatives, and other additives. These techniques allowed large-scale production and manufacturing of massively produced processed foods, making them available throughout the year. These processed products are time-saving as they require less preparation than fresh products—something that has become increasingly valued in high-income societies over the past few decades. Processed foods are also microbiologically safe, widely accessible and affordable to a growing population, and can contribute to food waste reduction as they have a longer shelf life than fresh foods. Moreover, some industrial processes—e.g., those used in tomato sauce preparation—might be beneficial as they could lead to enhanced bioaccessibility of antioxidants.⁸

In the last decades there is growing interest from the scientific community and especially medical experts in understanding and quantifying the risk factors related to dietary habits. Although this massive shift from artisanal food to processed products occurred in high-income countries first, middle-income and low-income countries have followed suit. Highly processed foods started to occupy supermarket shelves worldwide at the same time as a rise in the incidence of chronic inflammatory diseases, such as metabolic syndrome and inflammatory bowel disease. This correlation has led scientists worldwide to start investigating whether a link exists between this increasing degree of food processing and the risk of chronic diseases. In the last decades were carried a number of prospective epidemiological studies linking UPFs to human health, and describe recent mechanistic studies linking UPFs to chronic diseases, focusing on the role played by the intestinal microbiota.⁹

Ultra-processed food (UPF) consumption has been increasing in developed and developing countries in the last decades. Inevitably UPF has been the focus of major prospective epidemiological and clinical studies due to their potential negative effects on human health. UPFs have been suggested to impact the overall diet quality of populations globally. Due to the nature of their ingredients, UPFs represent an unhealthy dietary alternative growingly preferred among the younger generations.

Processed and ultra-processed foods are now critical risk factors to human health

Thanks to the implementation of nationwide comprehensive tobacco control initiatives in the last decades in most countries, cigarette smoking among adults has been declining steadily. In 2020, the most recent year for which such data are available, 12.5 % of U.S. adults age 18 and older smoked cigarettes, a significant decline from 42.4% of adults in 1965.

Tobacco continues to kill millions of people at premature age in Europe and globally. But smoking in European countries dropped substantially in the last decades, from 35-40% to 19% in recent years. Smoking in the EU in 2019 ranged from 6.4 % in Sweden to 28.7 % in Bulgaria.

Tobacco smoking cessation and control worldwide is associated with avoidance of millions of premature deaths, and an estimated extended mean lifespan of 19–20 years. Now Processed and ultra-processed foods are leading adverse health risk factors and causes of premature deaths.

In recent years, dozens of epidemiological studies have found that people who consume a lot of ultra-processed foods have higher rates of obesity, cardiovascular diseases (CVDs), type 2 diabetes, cancer and other chronic inflammatory diseases. The most important factors in Ultra-processed foods often are high levels of saturated fats, and high amounts of added salt and sugar. Studies also suggest that some additives in these foods could be responsible for negative health effects. The actual processing of the food could also make a difference to how human bodies respond to it. Studies have shown, for example, that when foods such as nuts are eaten whole the

body absorbs less of the fat than when the nut is ground down and the oils are released. Diets higher in UPFs could also affect human's gut health. Several studies have shown that eating higher amounts of UPFs is linked to a greater risk of high blood pressure, initiation of cardiovascular disease, heart attacks and strokes and early death.

The culmination of all investigative epidemiological evidence shows that prolonged exposure to processed and ultra-processed foods are associated with an increased risk of 32 damaging health outcomes including cancer, major heart and lung conditions, mental health disorders, and early death. The most recent findings were published in *The British Medical Journal* (BMJ, 28.2.2024). The paper contained systematic reviews and meta-analyses of cohort, case-control, and/or cross sectional study designs. The search identified 45 unique pooled analyses, including 13 dose-response associations and 32 non-dose-response associations (number of participants, n=9 888 373). Overall, direct associations were found between exposure to ultra-processed foods (UPFs) and 32 (71%) health parameters spanning mortality, cancer, and mental, respiratory, cardiovascular, gastrointestinal, and metabolic health outcomes.

Lane MM, Gamage E, Du S, Ashtree DN, et al. Ultra-processed food exposure and adverse health outcomes: umbrella review of epidemiological meta-analyses. *British Medical Journal* 2024; 384, e077310. DOI: <https://doi.org/10.1136/bmj-2023-077310> (28 February 2024).

Ultra-processed food (UPF) consumption has been increasing worldwide

Traditional dietary patterns based on the consumption of unprocessed or minimally processed foods, such as those typical of the Mediterranean area, are strongly associated with a lower risk for the onset and development of several non-communicable diseases (NCDs), such as obesity, types 2 diabetes, cardiovascular diseases (CVD) and some types of cancer, which represent the main causes of disability and premature death worldwide.^{10,11,12,13}

UPF consumption has been increasing worldwide in both adults and children over the past few decades.^{14,15,16} For example, representative

studies published since 2016 have shown that UPFs contributed up to 31% of daily caloric intake in France,¹⁷ 56 % in the UK,¹⁸ and 58 % in the USA.¹⁹ Consequently, exposure to numerous food additives has followed the same upwards trend, with the likes of dietary emulsifiers used in foods and beverages seeing annual sales constantly increasing.²⁰

Emulsifiers are added to processed foods like mayonnaise, ice cream, chocolates, peanut butter, cookies, creamy sauces, margarine, and baked goods to prevent the separation of their oil and water components. Emulsifiers also give these foods a smooth texture and increase their shelf life.

Numerous epidemiological and clinical studies have shown a correlation between ultra-processed food (UPF) consumption and the prevalence or incidence of numerous chronic inflammatory diseases [e.g., metabolic syndrome, IBD (Inflammatory Bowel Disease), and various cancers.

A recent review (2022) collected results from 48 prospective studies investigating the associations between UPF consumption and health status, chronic diseases, or mortality.²¹

Reviewers concluded that UPF consumption is associated with several adverse health effects and pathologies, which have serious epidemiological implications given the prevalence of UPFs in diets globally. Numerous research efforts have focused on better understanding the mechanisms behind UPF's effects on health. These efforts have brought to light the potential **central role played by the intestinal microbiota** in connecting UPF and host health. The human intestinal microbiota consists of trillions of microorganisms that have co-evolved with their host for millennia and now perform core functions, such as food digestion and maturation of host immunity and metabolism. A combination of laboratory-based preclinical research, epidemiological studies, and clinical trials suggest that UPFs affect human health through alterations of the intestinal microbiota composition and function.

Seminal works from Jeffrey Gordon's group (Washington University, School of Medicine in St Louis, USA), published in 2006–08, showed microbiota composition alterations in murine models of obesity and in individuals with obesity. These studies also showed that microbiota transplantation from mice with diet-induced obesity to lean germ-free

recipients on a healthy diet was sufficient to promote fat deposition in the recipient animals.²²

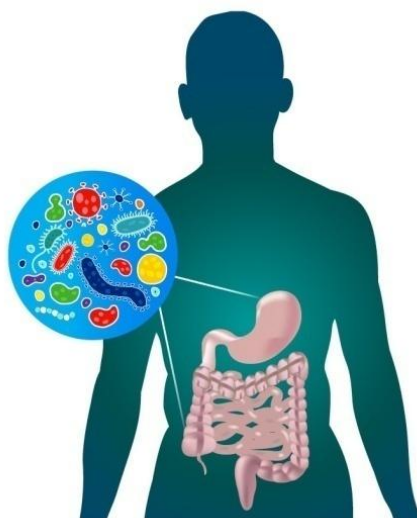


Figure 2. Gut microbiota, or gut flora are the microorganisms, including bacteria, archaea, fungi, and viruses, that live in the digestive tracts of humans. The human gastrointestinal tract (GIT) is composed of a variety of organs, each with varying internal environments. The study of chemical microbial products has grown into a field of research called microbial metabolomics. Intestinal microbiota produce a variety of compounds that are used by the body in biochemical reactions.

This scientific research established a causal link between diet-induced alterations of microbiota and altered host metabolism in mice. Mechanistically, functional microbiota analysis revealed an increase in features, such as import and processing of simple sugars by members of the microbiota in the presence of a high-fat diet, suggesting that the microbiota's ability to extract calories from ingested food affects diet-induced metabolic dysregulation. Another hallmark of metabolic dysregulation is the presence of chronic low-grade inflammation and alterations in microbiota composition and function, which lead to an increase in the systemic concentrations of bacterial products, such as lipopolysaccharides, derived from the outer cell membrane of Gram-negative bacteria contain a molecular structure known as lipid A and are able to cross the gastrointestinal mucosa via various mechanisms, ultimately reaching the systemic circulation. Once in the circulation, lipopolysaccharides infiltrate tissues (e.g., the liver and adipose tissues) and **trigger**

inflammation-related processes that associate with metabolic pathologies, such as obesity and insulin resistance.

The concept of **metainflammation**, a metabolic inflammatory state defined by low-grade chronic inflammation created by metabolic cells and stress sensors. Metainflammation was shown to contribute to obesity and insulin resistance over time in both mice and humans, through the infiltration of immune cells and the secretion of inflammatory cytokines in the tissue environment, which progressively interfered with—and even inhibited—insulin signalling. The demonstration that the immune system and metabolism were connected opened a new field of study in metabolic diseases, ultimately leading to the investigation of anti-inflammatory therapies as a treatment for obesity and metabolic diseases.²³

Moreover, intestinal microbiota are also highly regulated in their ability to produce metabolites, mainly through modulation of gene expression by the intestinal environment, including diet. Hence scientists realized that, UPFs also have the capacity to alter the metabolomic capacities of the intestinal microbiota in a way that can substantially affect host health.²⁴

For example, both human and animal studies have shown that microbiota members can respond to compounds present in UPFs by increasing their expression of virulence factors, consequently increasing the inflammatory potential of the microbiome. UPFs might also contribute to chronic inflammation by altering the production of beneficial bacterial metabolites such as short-chain fatty acids by the intestinal microbiota. Another example that highlights the importance of UPF–microbiota interactions in various chronic diseases relates to the increased consumption of refined sugar. In particular, scientists showed that the administration of a high-fructose diet to mice induced a typical microbiota encroachment to the gut epithelium, reduced mucus layer thickness, and altered intestinal microbiota composition. Scientists reported intestinal microbiota dysbiosis in mice consuming a high-fat and high-sucrose diet, which was associated with a predisposition to dextran sulfate sodium-induced colitis. Thus, UPFs disturb numerous mechanisms that are important for the maintenance of energy balance and immune homeostasis, ultimately leading to metabolic and inflammatory diseases.²⁵

Consumption of UPFs and health. NutriNet-Santé study

One of the most comprehensive tools used to measure UPF consumption is a set of detailed and repeated 24-h dietary records, especially when merged with extensive food composition databases. This tool also offers the opportunity to collect data about specific consumed brands within one generic food item, allowing for a more accurate UPF categorisation. When 24-h dietary records are not available, food frequency questionnaires, with a large number of items, can also be used to measure UPFs in diet; however, they are less accurate than 24-h dietary records. The largest number of prospective studies addressing the relationship between UPFs and health was conducted using the French NutriNet-Santé cohort (using repeated 24-h dietary records)²⁶

This study was aiming to assess the prospective associations between consumption of ultra-processed food and risk of cancer. There were 104,980 participants aged at least 18 years (median age 42.8 years) from the French NutriNet-Santé cohort (2009-17). Dietary intakes were collected using repeated 24 hour dietary records, designed to register participants' usual consumption for 3,300 different food items. These were categorised according to their degree of processing by the NOVA classification. The NOVA classification system groups all foods according to the nature, extent and purposes of the industrial processes they undergo. The results were focus on the associations between ultra-processed food (UPF) intake and risk of overall, breast, prostate, and colorectal cancer assessed by multivariable Cox proportional hazard models adjusted for known risk factors.

Results: Ultra-processed food (UPF) intake was associated with higher overall cancer risk (n= 2,228 cases; hazard ratio for a 10% increment in the proportion of ultra-processed food in the diet 1.12 (95% confidence interval 1.06 to 1.18) and breast cancer risk (n=739 cases; hazard ratio 1.11 (1.02 to 1.22). These results remained statistically significant after adjustment for several markers of the nutritional quality of the diet (lipid, sodium, and carbohydrate intakes and/or a Western pattern derived by principal component analysis).

The researchers concluded that in this large prospective study, a 10% increase in the proportion of ultra-processed foods in the diet was associated with a significant increase of greater than 10% in risks of overall and breast cancer. Further studies are needed to better understand the relative effect of the various dimensions of processing (nutritional composition, food additives, contact materials, and neoformed contaminants) in these associations.²⁶

Association of PF and UPF consumption and human health outcomes. From epidemiology to mechanistic insights

Numerous studies have shown a correlation between UPF consumption and the prevalence or incidence of numerous chronic inflammatory diseases (e.g., metabolic syndrome, Inflammatory Bowel Disease, and various digestive cancers). A very extensive recent review (2022) collected data from **48 prospective (epidemiological) studies** investigating the associations between UPF consumption and health status, chronic diseases, or mortality have been published. One of the most comprehensive tool used to measure UPF consumption is a set of detailed and repeated 24-h dietary records, especially when merged with extensive food composition databases. This tool also offers the opportunity to collect data about specific consumed brands within one generic food item, allowing for a more accurate UPF categorisation. When 24-h dietary records are not available, food frequency questionnaires, with a large number of items, can also be used to measure UPFs in diet; however, they are less accurate than 24-h dietary records. The largest number of prospective studies addressing the relationship between UPFs and health was conducted using the French NutriNet-Santé cohort (using repeated 24-h dietary records) and the Spanish SUN cohort (using a 136-item food frequency questionnaire), and some of these studies were summarised in six meta-analyses.

[Srouf B, Kordahi MC, Bonazzi E, et al. Ultra-processed food and human health: from epidemiological evidence to mechanistic insights. Review. *The Lancet Gastroenterology & Hepatology* 7(12): P1128-1140, 2022.]

The 2022 review examined the data in detail of the 48 prospective studies, of which 6 (six) have data from the general population and 1 (one) in

patients with cardiovascular disease (CVD). All these studies were conducted in Spain, France, Italy, and the USA. Their results suggested an association between consumption of UPFs and risk of all-cause mortality, with hazard ratios (HRs) ranging from 1·26 to 1·62 for the highest versus lowest UPF consumption categories.

With regards to cardiovascular and cerebrovascular disease (CVD) and mortality, 5 (five) studies investigated the potential link between UPF consumption and cardiovascular disease and mortality, in both the USA and Europe, and showed a significant increase in risk. In 4 (four) European studies (namely in the NutriNet-Santé, SUN, UK Biobank, and Lifelines cohorts) have shown direct associations between UPF consumption and the risk of type 2 diabetes, with HRs ranging from 1·26 to 1·53 and an odds ratio (OR) of 1·80.

In 1 (one) prospective study researchers found a significant association between UPF consumption and the risk of gestational diabetes in a Spanish population. Also, a Brazilian epidemiological study suggested an association between consumption of UPF during pregnancy and gestational weight gain. Additionally, 4 (four) prospective studies with data from Brazil, Spain, and Mexico explored the associations between UPF and the risk of hypertension. Although 3 (three) of these studies (done in men and women) found direct significant associations of UPFs with hypertension. The association measures (i.e., hazard ratio, HRs, Odds ratio, ORs, or risk ratios, RR) ranging between 1·21 and 1·35. The Mexican study (contacted only in women) found no significant association. The magnitude of all the aforementioned association measure estimates are similar to that typically observed in nutritional epidemiology studies, reflecting a substantial number of additional individuals with chronic diseases and premature deaths.

All these studies were performed and published in the period 2018-2022. Only 1 (one) prospective cohort study, conducted within the NutriNet-Santé cohort, explored the association between UPF and cancer risk, showing an association with overall cancer risk and breast cancer risk. Several prospective epidemiological studies have observed associations between UPF consumption and increased risks of overweight or obesity (or both). The summarised odds ratios (Ors) from the most recent meta-analysis

published in 2021 were 1·55 for obesity, 1·36 for overweight, and 1·41 for abdominal obesity. Odds ratios (Or) frequently are used to present strength of association between risk factors and outcomes in the clinical and epidemiological scientific published research.

Several prospective epidemiological studies with data from children have shown associations between increased consumption of UPFs and increases in weight, waist circumference, fat-mass index, and worse lipid profiles. All these results showed accumulating evidence suggesting an association between UPF consumption and various debilitating chronic inflammatory diseases.

Also, data from the 2021 multicentric international Prospective Urban Rural Epidemiology (PURE) cohort showed an association between UPF consumption and an increased risk of IBD (inflammatory Bowel Disease), and a follow-up study showed that these associations were significant for Crohn's disease but not for ulcerative colitis. There was no evidence for a significant association between UPF and IBD in the French NutriNet-Santé cohort, but there were few IBD cases available for analysis at the time of the study. Other prospective epidemiological studies have observed associations between consumption of UPFs and increased risk of depression or depressive symptoms, dyslipidaemia, renal function decline, frailty, decreased grip strength, and non-alcoholic fatty liver disease.

The 2022 review [Srouf B, Kordahi MC, Bonazzi E, et al. Ultra-processed food and human health: from...2016 to...2022] of all 48 studies are referenced in the reference section from no 27-100.²⁷⁻¹⁰⁰

The review contains epidemiological studies that suggested a role for UPFs in numerous chronic inflammatory diseases such as inflammatory bowel diseases and metabolic syndrome. Preclinical and clinical studies are accumulating to better decipher the effects of various aspects of food processing and formulation on the aetiology of chronic, debilitating inflammatory diseases. The review provides an overview of the current scientific data that highlight an association between UPFs consumption and various chronic diseases, with a focus on mechanistic insights involving the intestinal microbiota.

UPF and chronic diseases: mechanistic insights

After many decades of research a consensus has emerged among scientists regarding the associations between UPF consumption and the incidence of various adverse health chronic conditions. A UPF-rich diet could affect human health by a number of mechanisms:

- a. **UPFs consumption lower nutritional quality of the food:** Increased amounts of salt, sugar, saturated fats and energy, and decreased amounts of fibre, vitamins and useful micronutrients are negative effects.
- b. **UPFs can produce changes in food matrix and texture:** this will have impact on digestibility, bioavailability of nutrients, duration of chewing, increased eating rate, and over consumption of food and energy leading to obesity,
- c. **UPFs can change contact materials :** these materials are mostly contaminants from packaging materials (e.g. bisphenols, phthalates, and mineral oils, microplastics, etc),
- d. **Contaminants from processing the UPFs:** Acrolein, furans, acrylamide (cooking at high temperature, industrial trans fatty acids, and activated glycation. Frying at high temperature potatoes causes the highest acrylamide formation. Roasting potato pieces causes less acrylamide formation, followed by baking whole potatoes. Boiling potatoes and microwaving whole potatoes with skin on to make “microwaved baked potatoes” does not produce acrylamide. [Based on Food and Drug Agency (FDA) studies.]
- e. **Food additives and other industrial ingredients:** colouring agents, emulsifiers, flavouring agents, corn syrup, modified starches and maltodextrin.

The latest study (2024) on UPFs linked to 32 harmful effects to human health

In February 2024 an extensive umbrella review was published in the British Medical Journal (BMJ) about UPF consumption and 32 harmful effects on human health, including a higher risk of heart disease, cancer, type 2 diabetes, adverse mental health and early death, according to the world's largest review of its kind. The findings come amid rapidly rising global consumption of UPF such as cereals, protein bars, fizzy drinks, ready meals and fast food. For example, in the UK and US, more than 50% of the average diet now consists of ultra-processed foods. Especially for young people who are poorer or come from disadvantaged areas and low income countries, a diet comprising as much as 80% UPF is typical. The results of this latest 2024, review involved almost 10 million people

[Lane MM, Gamage E, McGuinness A, et al. Ultra-processed food exposure and adverse health outcomes: umbrella review of epidemiological meta-analyses. *British Medical Journal* 2024; 384 doi: <https://doi.org/10.1136/bmj-2023-077310>]. (Published 28 February 2024)Cite this as: BMJ 2024;384:e077310 .

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10. School of Biotechnology, Dublin City University, Dublin, Ireland

The Data sources are: MEDLINE, PsycINFO, Embase, and the Cochrane Database of Systematic Reviews, as well as manual searches of reference lists from 2009 to June 2023.

Eligibility criteria for selecting studies: Systematic reviews and meta-analyses of cohort, case-control, and/or cross sectional study designs. To evaluate the credibility of evidence, pre-specified evidence classification criteria were applied, graded as convincing (“class I”), highly suggestive (“class II”), suggestive (“class III”), weak (“class IV”), or no evidence (“class V”). The quality of evidence was assessed using the GRADE (Grading of Recommendations, Assessment, Development, and Evaluations) framework, categorised as “high,” “moderate,” “low,” or “very low” quality.

Results. The search identified 45 unique pooled analyses, including 13 dose-response associations and 32 non-dose-response associations, approximately 10 million (n=9 888 373). Overall, direct associations were found between exposure to ultra-processed foods (UPFs) and 32 (71%) health parameters spanning mortality, cancer, and mental, respiratory, cardiovascular diseases (CVDs), gastrointestinal, and metabolic health outcomes. Based on the pre-specified evidence classification criteria, convincing evidence (class I) supported direct associations between greater ultra-processed food exposure and higher risks of incident cardiovascular disease related mortality (risk ratio 1.50, 95% confidence interval 1.37 to 1.63; GRADE=very low) and type 2 diabetes (dose-response risk ratio 1.12, 1.11 to 1.13; moderate), as well as higher risks of prevalent anxiety outcomes (odds ratio 1.48, 1.37 to 1.59; low) and combined common mental disorder outcomes (odds ratio 1.53, 1.43 to 1.63; low). Highly suggestive (class II) evidence indicated that greater exposure to ultra-processed foods was directly associated with higher risks of incident **all cause mortality** (risk ratio 1.21, 1.15 to 1.27; low), **heart disease related mortality** (hazard ratio 1.66, 1.51 to 1.84; low), **type 2 diabetes** (odds ratio 1.40, 1.23 to 1.59; very low), and **depressive outcomes** (hazard ratio 1.22, 1.16 to 1.28; low), together with higher risks of prevalent **adverse sleep related outcomes** (odds ratio 1.41, 1.24 to 1.61; low), wheezing (risk ratio 1.40, 1.27 to 1.55; low), and obesity (odds ratio 1.55, 1.36 to 1.77; low). Of the remaining 34 pooled analyses, 21 were graded as suggestive or weak strength (class III-IV) and 13 were graded

as no evidence (class V). Overall, using the GRADE framework, 22 pooled analyses were rated as low quality, with 19 rated as very low quality and four rated as moderate quality.

Reviewers **concluded** that exposure to UPFs was associated with higher risk of adverse health outcomes, especially cardiometabolic, common mental disorder, and mortality outcomes.

“... Although several meta-analyses have made efforts to consolidate the many individual original research articles that have investigated the associations between exposure to UPFs and the risk of adverse health outcomes in the past decade, no comprehensive umbrella review has offered a broad overview and assessment of the existing meta-analytic evidence. Undertaking such a comprehensive review has the potential to enhance our understanding of these associations and provide valuable insights for better informing public health policies and strategies. This is particularly pertinent as the global debate continues regarding the need for public health measures to tackle exposure to ultra-processed foods in general populations. To bridge this gap in evidence and contribute to the ongoing discussion on the role of uUPFs exposure in chronic diseases, we did an umbrella review to evaluate the evidence provided by meta-analyses of observational epidemiological studies exploring the associations between exposure to ultra-processed food and the risk of adverse health outcomes....”.

RESULTS

Cancer

Pooled analyses from seven cohort studies showed direct associations between greater exposure to ultra-processed foods and higher risks of incident cancer overall (hazard ratio 1.12, 1.06 to 1.19; class III; very low).

Mental health

Examining data from two to four cross sectional designs, we found evidence supporting direct associations between greater exposure to ultra-processed foods and a higher risk of the prevalence of adverse sleep related outcomes (odds ratio 1.41, 1.24 to 1.61; class II; low)

Respiratory health

Pooled analyses that included two cross sectional studies provided limited evidence of an association between greater exposure to ultra-processed foods and risks of prevalent asthma (risk ratio 1.20, 0.99 to 1.46; class V; very low) and wheezing (risk ratio 1.40, 1.27 to 1.55; class II; low).

Cardiovascular health

Pooled analyses from six cohorts showed direct associations between greater ultra-processed food exposure and higher risks of incident cardiovascular

disease events such as morbidity and mortality (dose-response risk ratio 1.04, 1.02 to 1.06; class III; low and non-dose-response risk ratio 1.35, 1.18 to 1.54; class III; very low),

Gastrointestinal health

We found weak or no evidence in pooled analyses incorporating data from four cohorts for associations between greater exposure to ultra-processed foods and higher risks of incident Crohn's disease (hazard ratio 1.71, 1.37 to 2.14; class IV; low) and ulcerative colitis (hazard ratio 1.17, 0.86 to 1.61; class V; very low).

Metabolic health

The risk of abdominal obesity was examined by synthesising effect estimates from mixed cohort and cross sectional study designs, which showed direct associations with greater ultra-processed food exposure (dose-response odds ratio 1.05, 1.02 to 1.07; class III; low and non-dose-response odds ratio 1.41, 1.18 to 1.68; class III; very low).. We found weak to no evidence for associations between exposure to ultra-processed foods and hyperglycaemia (odds ratio 1.10, 0.34 to 3.52; class V; very low),

Conclusions

This umbrella review (2024) reports a higher risk of adverse health outcomes associated with UPFs exposure. The strongest available evidence pertained to direct associations between greater exposure to ultra-processed foods and higher risks of all cause mortality, cardiovascular disease related mortality, common mental disorder outcomes, overweight and obesity, and type 2 diabetes. Evidence for the associations of ultra-processed food exposure with asthma, gastrointestinal health, some cancers, and intermediate cardiometabolic risk factors remains limited and warrants further investigation.

References

1. Rumiati RI, Foroni F. We are what we eat: how food is represented in our mind/brain. *Psychonomic Bulletin & Review* 23:1043-1054, 2016.
2. Esch T, Kim JW, Stefano GB. Neurological implications of eating healthy and its association with pleasure. *Neuroendocrinology Letters* 27(1-2): 21-33, 2006.
3. Kringelbach ML, Stein A, van Hartevelt T. The functional human neuroanatomy of food pleasure cycles. Review. *Physiology & Behavior* 106(3):307-316, 2012.
4. Harvard T.H. Chan School of Public Health, The Nutrition Source, August 2023, [<https://www.hsph.harvard.edu/nutritionsource/processed-foods/>].
5. Mertens E, Collizzi C, Penalvo JL. Ultra-processed food consumption in adults across Europe. *European J Nutrition* 61:1521-1539, 2022.
6. Medical News Today, How do processed food affect your health, <https://www.medicalnewstoday.com/articles/318630>].
7. Lian Y, Wang G-P, Chen G-Q, Chen H-N. Association between ultra-processed foods and risk of cancer: a systematic review and meta-analysis. *Frontiers of Nutrition* 10:1175994, 2023. doi: 10.3389/fnut.2023.1175994.
8. Jacob HE. Six Thousand Years of Bread: its Holy and Unholy History. Lyons & Burford books, New York, New York, 1944.
9. Tomas M, Beekwilder J, Hall RD, et al. Industrial processing versus home processing of tomato sauce: effects on phenolics, flavonoids and in vitro bioaccessibility of antioxidants. *Food Chemistry* 220: 51-58, 2017.
10. Srour B, Kordahi M, Bonazzi E, et al. Ultra-processed foods and human health: from epidemiological evidence to mechanistic insights. *The Lancet Gastroenterology and Hepatology* 7(12): P1128-P1140, 2022, DOI: [https://doi.org/10.1016/S2468-1253\(22\)00169-8](https://doi.org/10.1016/S2468-1253(22)00169-8).
11. Dominguez LJ, Di Bella G, Veronese N, Barbagallo M. Impact of Mediterranean Diet on chronic non-communicable diseases and longevity. *Nutrients*. 13: 2028, 2021.
12. Sofi F, Abbate R, Gensini GF, Casini A. Accruing evidence on benefits of adherence to the Mediterranean diet on health: An updated systematic review and meta-analysis. *American J. Clin. Nutr* 92:1189–1196, 2010.
13. Lim SS, Vos T, Flaxman AD, et al. A comparative risk assessment of burden of disease and injury attributable to 67 risk factors and risk factor clusters in 21 regions, 1990-2010: A systematic analysis for the Global Burden of Disease Study 2010. *Lancet* 380: 2224–2260, 2010.
14. Ng M, Fleming T, Robinson M, et al. Global, regional, and national prevalence of overweight and obesity in children and adults during 1980-2013: A systematic analysis for the Global Burden of Disease Study 2013. *Lancet* 384:766–781, 2014.
15. Wang L, Martínez Steele E, Du M, et al. Trends in consumption of ultraprocessed foods among US youths aged 2–19 years, 1999–2018. *JAMA* 326: 519-530, 2021.
16. Juul F, Hemmingsson E. Trends in consumption of ultra-processed foods and obesity in Sweden between 1960 and 2010. *Public Health Nutrition* 18: 3096-3107, 2015.

17. Moubarac JC, Batal M, Martins AP, et al. Processed and ultra-processed food products: consumption trends in Canada from 1938 to 2011. *Can J Diet Pract Research* 75: 15-21, 2014.
18. France. Calixto Andrade G, Julia C, Deschamps V, et al. Consumption of ultra-processed food and its association with sociodemographic characteristics and diet quality in a representative sample of french adults., *Nutrients* 13: 682-, 2021.
19. Rauber F, da Costa Louzada ML, Steele EM, et al. Ultra-processed food consumption and chronic non-communicable diseases-related dietary nutrient profile in the UK (2008–2014). *Nutrients* 10:587, 2018.
20. Martinez SE, Baraldi LG, Louzada ML, et al. Ultra-processed foods and added sugars in the US diet: evidence from a nationally representative cross-sectional study. *BMJ Open*. 6e009892, 2016.
21. Roberts CL, Rushworth SL, Richman E, Rhodes JM. Hypothesis: increased consumption of emulsifiers as an explanation for the rising incidence of Crohn's disease. *J Crohn's Colitis* 7: 338-341, 2013.
22. Srouf B, Kordahi C, Bonazzi E. Ultra-processed foods and human health: from epidemiological evidence to mechanistic insights. Review. *The Lancet Gastroenterology and Hepatology* 7(12):1128-1140, 2022.
Srouf B, Touvier M. Processed and ultra-processed foods: coming to a health problem? Editorial *International J Food Sciences and Nutrition* . 71(6): 653-655, 2020. [https://doi.org/ 10.1080/09637486.2020.1807476](https://doi.org/10.1080/09637486.2020.1807476).
23. Ley RE, Turnbaugh PJ, Klein S, Gordon JL. Microbial ecology: human gut microbes associated with obesity. *Nature* : 444: 1022-1023, 2006.
24. Gregor MF, Hotamisligil GS. Inflammatory mechanisms in obesity. *Annual Review of Immunology* 29: 415-445, 2011.
25. Zinöcker MK, Lindseth IA. The western diet-microbiome-host interaction and its role in metabolic disease. *Nutrients*, 10: 365, 2018.
26. Fiolet T, Srouf B, Sellem L, Kesse-Guyot E, et al. 2018. Consumption of ultra-processed foods and cancer risk: results from NutriNet-Santé prospective cohort. *British Medical Journal*. 360: k322 , 2018.
27. Sandoval-Insausti H, Jiménez-Onsurbe M, Donat-Vargas C, et al. Ultra-processed food consumption is associated with abdominal obesity: a prospective cohort study in older adults. *Nutrients* 12e2368, 2020.
28. Bonaccio M, Costanzo S, Di Castelnuovo A, et al. Ultra-processed food intake and all-cause and cause-specific mortality in individuals with cardiovascular disease: the Moli-sani Study. *European Heart J*. 43: 213-224, 2022.
29. Bonaccio M, Di Castelnuovo A, Costanzo S, et al. Ultra-processed food consumption is associated with increased risk of all-cause and cardiovascular mortality in the Moli-sani Study. *American J Clin Nutrition* 113: 446-455, 2021.
30. Kim H , Hu EA , Rebholz CM. Ultra-processed food intake and mortality in the USA: results from the Third National Health and Nutrition Examination Survey (NHANES III, 1988-1994). *Public Health Nutrition* 22: 1777-1785, 2019.
31. Schnabel L, Kesse-Guyot E, Allès B, et al. Association between ultraprocessed food consumption and risk of mortality among middle-aged adults in France. *JAMA Intern Medicine* 179: 490-498, 2019.

32. Rico-Campà A, Martínez-González MA, Alvarez-Alvarez I, et al. Association between consumption of ultra-processed foods and all cause mortality: SUN prospective cohort study. *British Medical Journal* 365l1949, 2019.
33. Romero Ferreiro C, Martín-Arriscado Arroba C, Cancelas Navia P, et al. Ultra-processed food intake and all-cause mortality: DRECE cohort study. *Public Health Nutrition* 2021; <https://doi.org/10.1017/S136.898.0021003256>.
34. Blanco-Rojo R, Sandoval-Insausti H, López-García E, et al. Consumption of ultra-processed foods and mortality: a national prospective cohort in Spain. *Mayo Clin Proc* 94: 2178-2188, 2019.
35. Melo B, Rezende L, Machado P, et al. Associations of ultra-processed food and drink products with asthma and wheezing among Brazilian adolescents. *Pediatr Allergy Immunology* 29: 504-511, 2018.
36. Machado Azeredo C, Cortese M, Costa CS, et al. Ultra-processed food consumption during childhood and asthma in adolescence: data from the 2004 Pelotas birth cohort study. *Pediatr Allergy Immunology* 31: 27-37, 2020.
37. Fiolet T, Srour B, Sellem L, et al. Consumption of ultra-processed foods and cancer risk: results from NutriNet-Santé prospective cohort. *BMJ*. 360: k322, 2018.
38. Srour B, Fezeu LK, Kesse-Guyot E, et al. Ultra-processed food intake and risk of cardiovascular disease: prospective cohort study (NutriNet-Santé). *BMJ*. 365l1451, 2019.
39. Zhong G-C, Gu H-T, Peng Y, et al. Association of ultra-processed food consumption with cardiovascular mortality in the US population: long-term results from a large prospective multicenter study. *Int J Behav Nutr Phys Act*. 18: 21, 2021.
40. Du S, Kim H, Rebholz CM. Higher ultra-processed food consumption is associated with increased risk of incident coronary artery disease in the atherosclerosis risk in communities study. *J Nutr*. 151: 3746-3754, 2021.
41. Lo C-H, Khandpur N, Rossato SL, et al. Ultra-processed foods and risk of Crohn's disease and ulcerative colitis: a prospective cohort study. *Clin Gastroenterol Hepatol*. 20: e1323-e1337, 2021.
42. Juul F, Vaidean G, Lin Y, et al. Ultra-processed foods and incident cardiovascular disease in the framingham offspring study. *J Am Coll Cardiology* 77: 1520-1531, 2021.
43. Gómez-Donoso C, Sánchez-Villegas A, Martínez-González MA, et al. Ultra-processed food consumption and the incidence of depression in a Mediterranean cohort: the SUN Project. *European J Nutrition* 59: 1093-1103, 2019.
44. Adjibade M, Julia C, Allès B, et al. Prospective association between ultra-processed food consumption and incident depressive symptoms in the French NutriNet-Santé cohort. *BMC Medicine*. 17: 78, 2019.
45. Donat-Vargas C, Sandoval-Insausti H, Rey-García J, et al. High consumption of ultra-processed food is associated with incident dyslipidemia: a prospective study of older adults. *J Nutrition* 151: 2390-2398, 2021.

46. Costa CS, Assunção MCF, Loret de Mola C, et al. Role of ultra-processed food in fat mass index between 6 and 11 years of age: a cohort study. *International J Epidemiology* 50: 256-265, 2021.
47. Sandoval-Insausti H, Blanco-Rojo R, Graciani A, et al. Ultra-processed food consumption and incident frailty: a prospective cohort study of older adults. *J Gerontol A Biol Sci Med Sci*. 75: 1126-1133, 2020.
48. Zhang S, Gu Y, Rayamajhi S, et al. Ultra-processed food intake is associated with grip strength decline in middle-aged and older adults: a prospective analysis of the TCLSIH study. *European J Nutrition* 61: 1331-1341, 2022.
49. Leone A, Martínez-González MÁ, Craig W, et al. Pre-gestational consumption of ultra-processed foods and risk of gestational diabetes in a Mediterranean cohort. The SUN project. *Nutrients* 132202, 2021.
50. Gomes CB, Malta MB, Benício MHD, Carvalhaes MABL Consumption of ultra-processed foods in the third gestational trimester and increased weight gain: a Brazilian cohort study. *Public Health Nutr*. 24: 3304-3312, 2021.
51. Scaranni PODS, Cardoso LO, Chor D, et al. Ultra-processed foods, changes in blood pressure and incidence of hypertension: the Brazilian Longitudinal Study of Adult Health (ELSA-Brasil). *Public Health Nutr*. 24: 3352-3360, 2021.
52. Mendonça RD, Lopes AC, Pimenta AM, et al. Ultra-processed food consumption and the incidence of hypertension in a Mediterranean cohort: the Seguimiento Universidad de Navarra Project. *American J Hypertension* 30: 358-366, 2017.
53. Rezende-Alves K, Hermsdorff HHM, Miranda AEDS Food processing and risk of hypertension: cohort of Universities of Minas Gerais, Brazil (CUME Project). *Public Health Nutrition* 24: 4071-4079, 2021.
54. Monge A, Silva Canella D, López-Olmedo N, et al. Ultraprocessed beverages and processed meats increase the incidence of hypertension in Mexican women. *British J Nutrition* 126: 600-611, 2021.
55. Vasseur P, Dugelay E, Benamouzig R, et al. Dietary patterns, ultra-processed food, and the risk of inflammatory bowel diseases in the NutriNet-Santé cohort. *Inflamm Bowel Disease* 27: 65-73, 2021.
56. Narula N, Wong ECL, Dehghan M, et al. Association of ultra-processed food intake with risk of inflammatory bowel disease: prospective cohort study. *BMJ*. 374n1554, 2021.
57. Leffa PS, Hoffman DJ, Rauber F, et al. Longitudinal associations between ultra-processed foods and blood lipids in childhood. *Br J Nutr*. 124: 341-348, 2020.
58. Rauber F, Campagnolo PD, Hoffman DJ, Vitolo MR. Consumption of ultra-processed food products and its effects on children's lipid profiles: a longitudinal study. *Nutr Metab Cardiovasc Dis*. 25: 116-122, 2015.
59. Zhang S, Gan S, Zhang Q, et al. Ultra-processed food consumption and the risk of non-alcoholic fatty liver disease in the Tianjin chronic low-grade systemic inflammation and health cohort study. *Int J Epidemiology* 51: 237-249, 2021.
60. Rauber F, Chang K, Vamos EP, et al. Ultra-processed food consumption and risk of obesity: a prospective cohort study of UK Biobank. *Eur J Nutr*. 60: 2169-2180, 2021.

61. de Melo JMM, Dourado BLLFS, de Menezes RCE, et al. Early onset of overweight among children from low-income families: the role of exclusive breastfeeding and maternal intake of ultra-processed food. *Pediatr Obes*. 16e12825, 2021.
62. Li M, Shi Z. Ultra-processed food consumption associated with overweight/obesity among Chinese adults-results from China health and nutrition survey 1997–2011. *Nutrients* 13:2796, 2021.
63. Canhada SL, Luft VC, Giatti L, et al. Ultra-processed foods, incident overweight and obesity, and longitudinal changes in weight and waist circumference: the Brazilian Longitudinal Study of Adult Health (ELSA-Brasil). *Public Health Nutr* 23: 1076-1086, 2020.
64. Beslay M, Srouf B, Méjean C, et al. Ultra-processed food intake in association with BMI change and risk of overweight and obesity: a prospective analysis of the French NutriNet-Santé cohort. *PLoS Med*, 17: Article e1003256, 2020.
65. Mendonça RD, Pimenta AM, Gea A, et al. Ultraprocessed food consumption and risk of overweight and obesity: the University of Navarra follow-up (SUN) cohort study. *Am J Clin Nutr* 104: 1433-1440, 2016.
66. Cordova R, Kliemann N, Huybrechts I, et al. Consumption of ultra-processed foods associated with weight gain and obesity in adults: a multi-national cohort study. *Clinical Nutrition* 40: 5079-5088, 2021.
67. Rey-García J, Donat-Vargas C, Sandoval-Insausti H, et al. Ultra-processed food consumption is associated with renal function decline in older adults: a prospective cohort study. *Nutrients* 13 : 428, 2021.
68. Srouf B, Fezeu LK, Kesse-Guyot E, et al. Ultraprocessed food consumption and risk of type 2 diabetes among participants of the NutriNet-Santé prospective cohort. *JAMA Intern Med* 180: 283-291, 2020.
69. Llaveró-Valero M, Escalada-San J, Martín MA, et al. Ultra-processed foods and type-2 diabetes risk in the SUN project: a prospective cohort study. *Clinical Nutrition* 40: 2817-2824, 2021.
70. Levy RB, Rauber F, Chang K, et al. Ultra-processed food consumption and type 2 diabetes incidence: a prospective cohort study. *Clinical Nutrition* 40: 3608-3614, 2020.
71. Duan M-J, Vinke PC, Navis G, et al.. Ultra-processed food and incident type 2 diabetes: studying the underlying consumption patterns to unravel the health effects of this heterogeneous food category in the prospective Lifelines cohort. *BMC Medicine* 20: 7-, 2022.
72. Konieczna J, Morey M, Abete I, et al. Contribution of ultra-processed foods in visceral fat deposition and other adiposity indicators: prospective analysis nested in the PREDIMED-Plus trial. *Clinical Nutrition* 40: 4290-4300, 2021.
73. Costa CS, Rauber F, Leffa PS, et al. Ultra-processed food consumption and its effects on anthropometric and glucose profile: a longitudinal study during childhood. *Nutr Metab Cardiovasc Dis* 29: 177-184, 2019.
74. Chang K, Khandpur N, Neri D, et al. Association between childhood consumption of ultraprocessed food and adiposity trajectories in the avon longitudinal study of parents and children birth cohort. *JAMA Pediatrics* 175: Article e211573, 2021.

75. Hercberg S, Castetbon K, Czernichow S, *et al.* The Nutrinet-Santé Study: a web-based prospective study on the relationship between nutrition and health and determinants of dietary patterns and nutritional status. *BMC Public Health* 10: 242, 2010.
76. Martínez-González MA. The SUN cohort study (Seguimiento University of Navarra). *Public Health Nutrition* 9: 127-131, 2006.
77. Lane MM, Davis JA, Beattie S, *et al.* Ultraprocessed food and chronic noncommunicable diseases: a systematic review and meta-analysis of 43 observational studies. *Obesity Reviews* 22: Article e13146, 2021.
78. Pagliai G, Dinu M, Madarena MP *et al.* Consumption of ultra-processed foods and health status: a systematic review and meta-analysis. *Br J Nutrition* 125: 308-318, 2021.
79. Askari M, Heshmati J, Shahinfar H, *et al.* Ultra-processed food and the risk of overweight and obesity: a systematic review and meta-analysis of observational studies. *Int J Obesity (London)*, 44: 2080-2091, 2020.
80. Moradi S, Entezari MH, Mohammadi H, *et al.* Ultra-processed food consumption and adult obesity risk: a systematic review and dose-response meta-analysis. *Critical Rev Food Sci Nutr*, 2021. <https://doi.org/10.1080/10408398.2021.1946005>, 2021. , and Moradi S, Entezari MH, Mohammadi H, *et al.* Ultra-processed food consumption and adult obesity risk: a systematic review and dose-response meta-analysis. Meta-analysis. *Critical Review Food Science Nutrition* 63(2):249-260, 2023. doi: [10.1080/10408398.2021.1946005](https://doi.org/10.1080/10408398.2021.1946005). Epub 2021 Jun 30.
81. Suksatan W, Morad S, Naeini F, *et al.* Ultra-processed food consumption and adult mortality risk: a systematic review and dose-response meta-analysis of 207,291 participants. *Nutrients* 14 : 174, 2021.
82. Delpino FM, Figueiredo LM, Bielemann RM, *et al.* Ultra-processed food and risk of type 2 diabetes: a systematic review and meta-analysis of longitudinal studies. *International Journal of Epidemiology* (2021). <https://doi.org/10.1093/ije/dyab247>.
83. Mendonca RD, Lopes AC, Pimenta AM, *et al.* Ultra-processed food consumption and the incidence of hypertension in a Mediterranean cohort: the Seguimiento Universidad de Navarra Project. *American Journal of Hypertension* 30 : 358-366, 2017.
84. Schwingshackl L, Schwedhelm C, Hoffmann G, *et al.* Food groups and risk of all-cause mortality: a systematic review and meta-analysis of prospective studies. *Am J Clinical Nutrition* 105: 1462-1473, 2017.
85. Afshin A, Sur PJ, Fay KA, *et al.* Health effects of dietary risks in 195 countries, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017, *Lancet* 393: 1958-1972, 2019.
86. Kliemann N, Al Nahas A, Vamos EP, *et al.* Ultra-processed foods and cancer risk: from global food systems to individual exposures and mechanisms. *British J Cancer* 127 (1): 14-20, 2022.
87. Zhang S, Gu Y, Rayamajhi S, *et al.* Ultra-processed food intake is associated with grip strength decline in middle-aged and older adults: a prospective analysis of the TCLSIH study. *European J Nutr* 61: 1331-1341, 2022.

88. Hall KD, Ayuketah A, Brychta R, *et al.* Ultra-processed diets cause excess calorie intake and weight gain: an inpatient randomized controlled trial of ad libitum food intake. *Cell Metab*, 30: 226, 2019.
89. Luiten CM, Steenhuis IH, Eyles H, *et al.* Ultra-processed foods have the worst nutrient profile, yet they are the most available packaged products in a sample of New Zealand supermarkets. *Public Health Nutr* 19: 539, 2016.
90. Adams J, White M. Characterisation of UK diets according to degree of food processing and associations with socio-demographics and obesity: cross-sectional analysis of UK National Diet and Nutrition Survey (2008–2012). *Int J Behav Nutr Phys Act* 12: 160, 2015.
91. Cediel G, Reyes M, da Costa Louzada ML, *et al.* Ultra-processed foods and added sugars in the Chilean diet (2010). *Public Health Nutr*, 21: 125-133, 2017.
92. Costa Louzad ML, Martins AP, Canella DS, *et al.* Ultra-processed foods and the nutritional dietary profile in Brazil. *Review Saude Publica* 49: 38, 2015.
93. Moubarac JC, Martins AP, Claro RM, *et al.* Consumption of ultra-processed foods and likely impact on human health. Evidence from Canada. *Public Health Nutrition* 16: 2240-2248, 2013.
94. Moubarac JC, Batal M, Louzada ML, *et al.* Consumption of ultra-processed foods predicts diet quality in Canada. *Appetite* 108 : 512-520, 2017.
95. Louzada ML, Martins AP, Canella DS, *et al.* Impact of ultra-processed foods on micronutrient content in the Brazilian diet. *Rev Saude Public*, 49: 45, 2015.
96. International Agency for Research Cancer. Nutri-Score: harmonized and mandatory front-of-pack nutrition label urgently needed at the European Union level and beyond, <https://www.iarc.who.int/news-events/nutri-score/> (Sept 1, 2021).
97. Mozaffarian D, Katan MB, Ascherio A, *t al.* Trans fatty acids and cardiovascular disease. *New Engl J Medicine* 354: 1601-1613, 2006.
98. EFSA (European Food Safety Authority) Panel on Contaminants in the Food Chain (CONTAM). Scientific opinion on acrylamide in food. EFSA J 13: Article 4104, 2015.
99. Martínez Steele E, Khandpur N, da Costa Louzada ML, Monteiro CA. Association between dietary contribution of ultra-processed foods and urinary concentrations of phthalates and bisphenol in a nationally representative sample of the US population aged 6 years and older. *PLoS One* 15: Article e0236738, 2020.
100. Buckley JP, Kim H, Wong E, Rebholz CM. Ultra-processed food consumption and exposure to phthalates and bisphenols in the US National Health and Nutrition Examination Survey, 2013–2014. *Environment International* October 131:105657, 2019. doi: DOI:10.1016/j.envint.2019.105057.